

A disproof of Conjecture 00000006120

Identical noise stability forces identical degree weights

Jackmeson1

1 The conjecture

Conjecture 00000006120 reads (English version):

Definition: Noise stability and the spectrum are two layers. *Conjecture:* There exist two Boolean functions with identical noise stability but different spectral distributions, and the separation is realized by an explicit perturbation with the same stability but different low-degree weights.

The Chinese version says that *all* noise stabilities of the two functions coincide (“yiqie zaosheng wendingxing xiangtong”), and that the separation is realized by an explicit perturbation with the same stability but different low-degree weights.

The claim is a conjunction: (A) two Boolean functions with identical noise stability, (B) with different spectral distributions, and (C) an explicit pair (“perturbation”) with the same stability but different low-degree weights. We show that (C) can never hold. The reason is that the function $\rho \mapsto \text{Stab}_\rho[f]$ is a polynomial whose coefficients are exactly the Fourier weights $W^k[f]$ at each degree k . So identical noise stability determines every degree weight, and hence every low-degree weight. (A) and (B) on their own are satisfiable (Section 4).

2 Definitions

All notions are the standard ones of O’Donnell’s *Analysis of Boolean Functions* (Cambridge University Press, 2014; arXiv:2105.10386), which we quote from <https://arxiv.org/html/2105.10386> (mathematical symbols are stripped in that rendering): “Definition 1.19 . For and , the (Fourier) weight of at degree is”; “the spectral sample for , denoted , is the probability distribution on subsets of in which the set has probability .”; “Definition 2.40 . Let . For fixed we write to denote that the random string is drawn as follows: for each independently,”; “Definition 2.42 . For and , the noise stability of at is”.

We encode the cube $\{-1, 1\}^n$ as Bool^n , with $\text{sgn}(\text{false}) = 1$ and $\text{sgn}(\text{true}) = -1$. All functions $f: \{-1, 1\}^n \rightarrow \mathbb{R}$ are allowed. A *Boolean function* takes only the values ± 1 (`IsBoolean`).

Definition 1. For $S \subseteq [n]$ let $\chi_S(x) = \prod_{i \in S} x_i$ (`chi`). The Fourier coefficient is $\hat{f}(S) = 2^{-n} \sum_x f(x) \chi_S(x)$ (`fourier`). The *spectral distribution* (spectral sample) gives the set S the weight $\hat{f}(S)^2$ (`spectralDist`). These weights sum to $\mathbf{E}[f^2]$, so they form a probability distribution exactly when $\mathbf{E}[f^2] = 1$, in particular for every Boolean (± 1 -valued) f ; for a general real-valued f we call this family the (unnormalized) spectral measure. The *Fourier weight at degree k* is $W^k[f] = \sum_{|S|=k} \hat{f}(S)^2$ (`weight`), and the *low-degree weight up to degree k* is $W^{\leq k}[f] = \sum_{|S| \leq k} \hat{f}(S)^2$ (`lowWeight`).

Definition 2. Let $\rho \in [-1, 1]$. The noise kernel is $K_\rho(x, y) = \prod_{i=1}^n [\frac{1+\rho}{2} \text{ if } y_i = x_i, \frac{1-\rho}{2} \text{ if } y_i \neq x_i]$ (`noiseKernel`). This is the law of $y \sim N_\rho(x)$: independently in each coordinate, $y_i = x_i$ with probability $\frac{1+\rho}{2}$ and $y_i = -x_i$ otherwise. The *noise stability* is $\text{Stab}_\rho[f] = \sum_x \sum_y 2^{-n} K_\rho(x, y) f(x) f(y) = \mathbf{E}[f(x)f(y)]$, where x is uniform and $y \sim N_\rho(x)$ (`stab`). The Lean definitions accept every real ρ ; outside $[-1, 1]$ the same formulas are only an algebraic extension (a kernel entry can be negative, e.g. $\frac{1-\rho}{2} = -\frac{1}{2}$ at $\rho = 2$), not a probability law or an expectation. The disproof only uses $\rho \in [0, 1]$.

As sanity checks, $K_\rho(x, \cdot)$ sums to 1 for every ρ (`noiseKernel_sum`) and is nonnegative for $\rho \in [-1, 1]$ (`noiseKernel_nonneg`).

3 The disproof

Lemma 3 (Fourier formula for noise stability). *For every $f: \{-1, 1\}^n \rightarrow \mathbb{R}$ and every real ρ , $\text{Stab}_\rho[f] = \sum_{S \subseteq [n]} \rho^{|S|} \hat{f}(S)^2 = \sum_{k=0}^n W^k[f] \rho^k$.*

This is the standard formula. Wikipedia (https://en.wikipedia.org/wiki/Analysis_of_Boolean_functions) says that the noise stability “can be defined in two equivalent ways”, namely $\text{Stab}_\rho[f] = \mathbf{E}_{x; y \sim N_\rho(x)}[f(x)f(y)] = \sum_{S \subseteq [n]} \rho^{|S|} \hat{f}(S)^2$. O’Donnell’s text has “Thus the noise stability of f is equal to the sum of its Fourier weights, attenuated by a factor which decreases exponentially with degree.” We prove the formula in Lean from the probabilistic definition above.

Proof. A case check gives $[\frac{1+\rho}{2} \text{ if } y_i = x_i \text{ else } \frac{1-\rho}{2}] = \frac{1}{2}(1 + \rho x_i y_i)$ (`factor_eq`). Expanding the product over subsets gives $K_\rho(x, y) = 2^{-n} \prod_i (1 + \rho x_i y_i) = 2^{-n} \sum_S \rho^{|S|} \chi_S(x) \chi_S(y)$ (`noiseKernel_expand`). Substituting and exchanging the finite sums gives $\text{Stab}_\rho[f] = \sum_S \rho^{|S|} (2^{-n} \sum_x f(x) \chi_S(x))^2$ (`stab_eq_fourier`). Grouping the sets by size gives the second form (`stab_eq_weights`). In Lean this is phrased with the polynomial $P_f = \sum_S \hat{f}(S)^2 X^{|S|} \in \mathbb{R}[X]$: we have $P_f(\rho) = \text{Stab}_\rho[f]$ (`eval_stabPoly`), and the k -th coefficient of P_f is $W^k[f]$ (`coeff_stabPoly`). $\square$

Theorem 4. *Let $f: \{-1, 1\}^n \rightarrow \mathbb{R}$ and $g: \{-1, 1\}^m \rightarrow \mathbb{R}$ (with n and m possibly different). Let $s \subseteq \mathbb{R}$ be infinite with $\text{Stab}_\rho[f] = \text{Stab}_\rho[g]$ for all $\rho \in s$. Then $W^k[f] = W^k[g]$ and $W^{\leq k}[f] = W^{\leq k}[g]$ for every $k \geq 0$.*

Proof. The polynomials P_f and P_g agree at infinitely many points, so $P_f = P_g$ (`Mathlib’s Polynomial.eq_of_infinite_eval_eq`). Comparing coefficients gives $W^k[f] = W^k[g]$ (`weight_eq_of_stab_eq`). Since $W^{\leq k} = \sum_{j \leq k} W^j$ (`lowWeight_eq_sum`), the low-degree weights agree too (`lowWeight_eq_of_stab_eq`). $\square$

Definition 5 (Realizing pair). A pair (f, g) *realizes the clause* (`RealizingPair`) if $\text{Stab}_\rho[f] = \text{Stab}_\rho[g]$ for every $\rho \in [0, 1]$, and either $W^k[f] \neq W^k[g]$ for some k , or $W^{\leq k}[f] \neq W^{\leq k}[g]$ for some k .

Requiring equality only on $[0, 1]$ is weaker than equality on all of $[-1, 1]$. So refuting this version also refutes the $[-1, 1]$ version.

Corollary 6. *No pair of real-valued functions on Boolean cubes realizes the clause (`no_realizing_pair`). Hence the conjecture is false (`conjecture6120_false`). Precisely, there are no n and no Boolean f, g on $\{-1, 1\}^n$ with $\text{Stab}_\rho[f] = \text{Stab}_\rho[g]$ for all $\rho \in [-1, 1]$ and $\hat{f}^2 \neq \hat{g}^2$ that come with Boolean f', g' (of any arities, possibly $f' = f$ and $g' = g$) forming a realizing pair (`Statement`).*

Proof. $[0, 1]$ is infinite, so Theorem 4 applies to any candidate realizing pair. $\square$

4 Non-vacuity

The first two clauses on their own are satisfiable, so the definitions are not degenerate. Noise stability is invariant under relabelling the variables (`stab_relabel`). The two dictators $x \mapsto x_0$ and $x \mapsto x_1$ on $\{-1, 1\}^2$ are Boolean, have $\text{Stab}_\rho[x_0] = \text{Stab}_\rho[x_1]$ for every real ρ , and have different spectral distributions, since $\hat{x}_0(\{0\})^2 = 1 \neq 0 = \hat{x}_1(\{0\})^2$ (`first_clauses_satisfiable`). By Theorem 4 they have the same W^k for every k , so they do not realize the clause.

5 Scope and readings

- **Refuted.** “Same stability” read as equality of Stab_ρ for all $\rho \in [-1, 1]$, for all $\rho \in [0, 1]$, or for any infinite set of ρ (Theorem 4). “Different low-degree weights” read as $W^k[f] \neq W^k[g]$ for some degree k , or as $W^{\leq k}[f] \neq W^{\leq k}[g]$ for some k . The realizing pair may be the separating pair itself or a different pair, and the two functions may have different numbers of variables. Booleanity is never used, so the result also holds verbatim for $\{0, 1\}$ -valued functions. “Perturbation” puts no further constraint on the pair, and adding one only makes the clause harder to satisfy.
- **Not refuted.** (i) Reading “low-degree weights” as the individual weights $\hat{f}(S)^2$ of single low-degree sets S . Under this reading the clause holds, for example for the dictators of Section 4. (ii) Reading “same stability” as equality at a single ρ (or at finitely many). At $\rho = 1$, every Boolean function has $\text{Stab}_1 = 1$. This reading contradicts “identical noise stability” (all noise stabilities, in the Chinese text).

6 Lean formalization

The file `lean/Conjecture6120/Basic.lean` (namespace `Conjecture6120`, Lean 4.33.1, Mathlib v4.33.1) contains every definition and result named above. Its main theorems are `conjecture6120_false` : $\neg \text{Statement}$, `no_realizing_pair`, `weight_eq_of_stab_eq`, `lowWeight_eq_of_stab_eq`, `stab_eq_fourier` and `first_clauses_satisfiable`. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for each of them, and also for `noiseKernel_sum`. The file contains no `sorry` and no `native_decide`.